Do not write outside the box

0 3 This question is about pollutants.

0 3 . 1 Waste water has harmful substances removed before being released into the environment.

Complete the sentences.

[2 marks]

Agricultural waste water requires the removal of harmful

_____ .

Industrial waste water may require the removal of harmful

_____ .

0 3 . 2 How is sewage sludge treated before being released into the environment?

[1 mark]

Tick (✓) **one** box.

Aerobic biological treatment ☐

Anaerobic digestion ☐

Grit removal ☐

Screening ☐

Do not write outside the box

0 3 . 3 Hydrocarbons are used to make polymers. Polymers are used to make plastic bags.

In one year 8.0 billion plastic bags were used.

The next year there was a charge for plastic bags and only 1.3 billion plastic bags were used.

Calculate the percentage decrease in the number of plastic bags used.

[3 marks]

_____

_____

_____

_____

_____

_____

Percentage decrease = _____ %

Question 3 continues on the next page

Do not write outside the box

Oxides of nitrogen are pollutants formed in car engines.

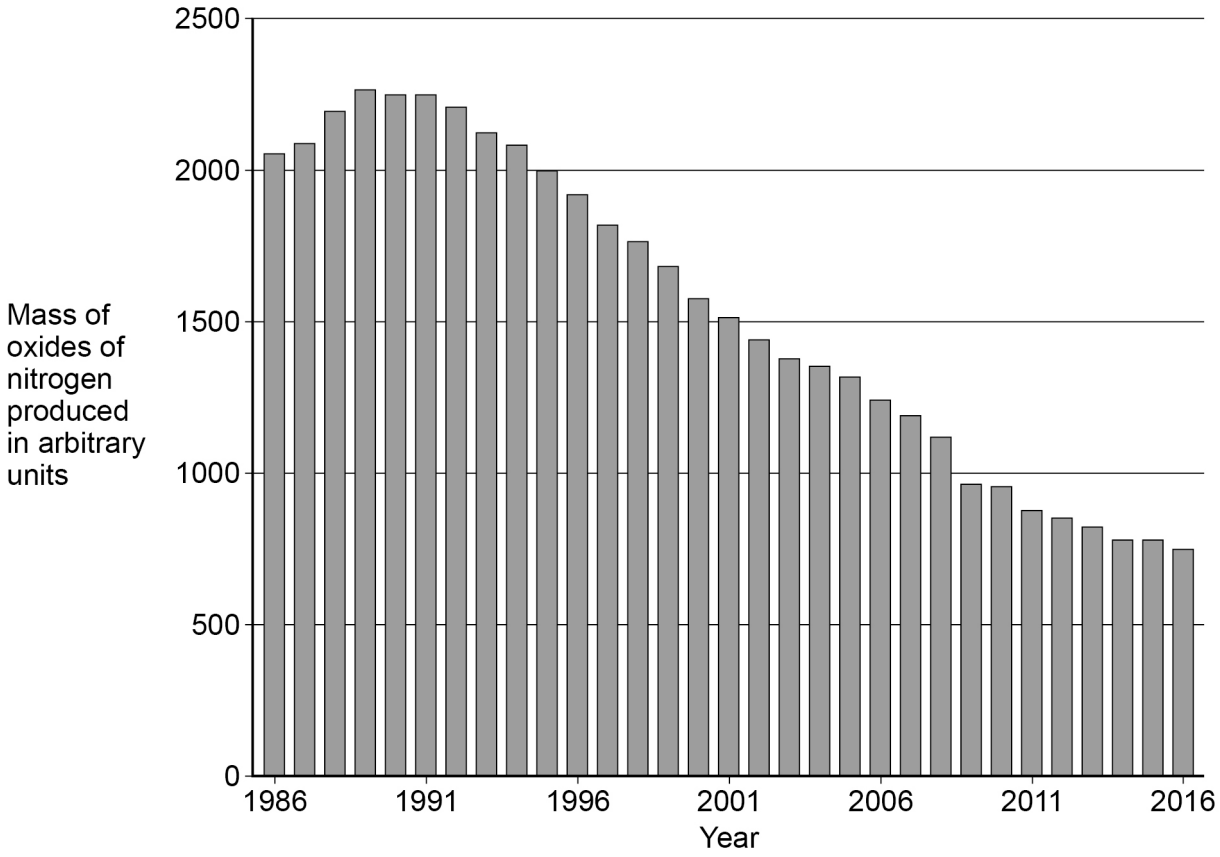
0 3 . 4 Give **one** problem oxides of nitrogen cause.

[1 mark]

_____

0 3 . 5 **Figure 3** shows the mass of oxides of nitrogen produced from car engines from 1986 to 2016.

Figure 3



Suggest why the mass of oxides of nitrogen produced from car engines increased and then decreased.

[2 marks]

Increased _____

_____

Decreased _____

_____

9

Do not write outside the box

Turn over for the next question

DO NOT WRITE ON THIS PAGE
ANSWER IN THE SPACES PROVIDED